structure in dense glassy mixtures is simple: it displays a broad continuous spectrum of geometrical arrangements, possibly decorated by features like LFS, crystallites or structural defects, or that the current identification of LFS is missing some crucial ingredient (e.g. information about dynamics). The outcome of PCA is also easy to interpret, as the principal component directions provide direct insight into the dominant structural parameters [57]. On the one hand, these results question the utility of complex deep learning methods in studying glass structure. On the other hand, the descriptors used in Refs. [12, 13, 51] do not exhaust all forms of structural heterogeneity and some may be affected by some deeper shortcomings [58]. Development of structural descriptors is still very active [59, 60] and these advances wait for applications in the context of glassy materials. Addressing the above issues may become crucial in future studies of more demanding benchmarks for structural characterization, such as compositional order in polydisperse glassy models [61, 62], medium-range order in oxides or metallic glasses [63, 64], and orientational order in glassy water [65, 66]. Computational studies of these complex systems represent exciting opportunities to gain insight into the nature and role of local structure in glassy materials and to provide solid grounds for predictive theoretical approaches based on structure.

Another research line where structure-based ML approaches are making progress aims at predicting macroscopic properties of glasses relevant for applications, such as oxides or chalcogenide glasses, over a wide range of chemical compositions [67, 68]. Recent work on sodium-silicate glasses shows that physics-informed machine learning models can reliably interpolate and extrapolate these properties on the basis of structural information only [69]. These findings indicate that, despite the apparent complexity of the feature space, the relationship between local structure and macroscopic glass properties is often linear, which makes it easy for machine learning models to generalize outside their training set. See Ref. [70] for a recent roadmap covering this topic.

Having characterized amorphous structure, a crucial question is whether the structural descriptors are connected to emergent relaxation dynamics in the glass-forming liquid [71–74]. As will be clear in Sec. V, current unsupervised methods provide only limited insights into the heterogeneity of the dynamics, except in specific systems dominated by strong icoshaedral order [12, 13, 51]. Whether this is a technical limitation of the unsupervised methods used to date or rather an intrinsic feature of supercooled dynamics remains to be clarified.

III. PREDICTION OF STRUCTURAL RELAXATION AND DYNAMIC HETEROGENEITIES

One of the central challenges for both computational and theoretical studies of glass-forming liquids is to predict future dynamics of a configuration from an initial snapshot. Note that one is not interested in predicting the whole future evolution, but only the dynamical processes leading to microscopic irreversible motion. Supervised ML provides a natural tool to perform such prediction, essentially by fitting high-dimensional structural input to the relaxation dynamics, similar in spirit to classification in image recognition. In general, three choices need to be made to design a model: i) which structural descriptors to use to characterize the input configuration; ii) which labels to use to quantify structural microscopic relaxation; and iii) what model and ML algorithm to use to fit the input to the labels. A large variety of techniques have already been introduced to tackle this problem, ranging from ridge regression using complex and coarse-grained structural descriptors to graph neural networks using raw particles positions, as illustrated in Fig. 3.

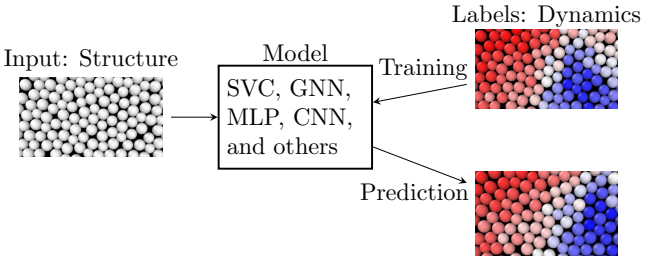

FIG. 3. Typical supervised ML procedure in condensed matter. The raw input is encoded using structural descriptors or graphs. A model is trained using labels which describe structural relaxation obtained, e.g., from molecular dynamics simulations. Colors indicate frozen (blue) or rearranging (red) particles. ML techniques range from support vector classification (SVC) to graph neural networks (GNNs), multilayer perceptrons (MLPs), and convolutional neural networks (CNNs). After training, structural relaxation is predicted for previously unseen structures.

In 2015, Cubuk *et al.* [20] demonstrated that support vector classifiers (SVC) [75] could be used to classify soft spots relaxing fast against slowly-relaxing regions in glasses. Here, soft spots are defined as regions that had a high likelihood of rearranging within a short time scale. As input to this algorithm, each particle was assigned a vector of local structural descriptors that captured the local density and angular structure within shells at different distances from its center. From the trained SVC a continuous structural descriptor called softness, S, can be extracted which correlates with the likelihood for the particle to rearrange in the near future. Softness has been used to gain insight into a variety of glass problems encompassing many different types of glassy liquids and disordered solids, ranging from strong to fragile and ductile to brittle with constituent particles ranging from atomic to granular, studied in bulk and in thin films [19, 76–84]. Further, this approach also led to a series of papers [31, 85, 86] aiming at the construction of an

effective model for the dynamics in glassy fluids built around the evolution of the softness field, which will be discussed in more detail in Sec. IV.

Whereas softness is associated to prediction on short time scales, several subsequent works have focused on predicting the dynamics, or more precisely the dynamic propensity [87], on longer time scales $t \sim \tau_\alpha$. Here, τ_α is the typical structural relaxation time scale on which each particle moves on average roughly one particle diameter (see SM [54]). Dynamic propensity quantifies the local dynamics of a glass-forming liquid by capturing the typical behaviour of each particle in a structure using the isoconfigurational ensemble [87]. This ensemble is formed by a set of trajectories which start from the same initial structure but with different initial velocities drawn from a Maxwell-Boltzmann distribution. By calculating the mean distance travelled by a given particle in this ensemble, one arrives at the dynamic propensity $\langle|\Delta\mathbf{r}_i(t)|\rangle_{\mathrm{iso}}$. Due to this averaging, the dynamic propensity captures the part of the dynamics encoded in the initial structure, leaving out the part stemming from the initial velocities which cannot be captured by any structural descriptor. While some dynamical information is necessarily lost [88], the dynamic propensity, which fluctuates from one particle to the other, is an important measure for dynamic heterogeneity in supercooled liquids. At the end of the section, we will also discuss ways to reintroduce the fluctuations around the isoconfigurational average using different labels or new ML designs.

In 2020, Bapst *et al.* [21] introduced a graph neural network (GNN) that could predict the dynamic propensity significantly better than a support vector machine (SVM) (where the input for the SVM was angular and radial functions similar to Ref. [20]). In contrast to the SVM, the input to the GNN was a graph structure where each particle in the initial configuration was represented by a vertex, and edges were drawn between particles within a cutoff radius of each other. In addition to the structure of this graph, the vertices and edges also carry information: the particle species (as vertex data) and the vectors connecting neighbours (as edge data). The GNN model then consists of several multilayer perceptrons (MLP) that iteratively update the features contained at the edges and nodes, with each iteration passing information along the nodes and edges of the network. After the final iteration, the features at the nodes are passed through a final MLP that predicts the dynamic propensity. As all vertices are updated in parallel, the network predicts mobilities of all particles simultaneously. This network was then optimized to minimize the squared difference between the predicted and true propensity using the $\mathcal{L}_2$ norm. Recently, two improvements to this GNN approach were proposed. Shiba *et al.* [25] demonstrated that a significantly higher accuracy could be reached over nearly all time scales by not only considering single particle dynamics but also pairwise dynamics. Specifically, they trained the GNN to predict not only the dynamic propensity but also the isoconfigurational change in the distance between pairs of particles sharing an edge in the graph — a modification called BOnd TArgeting Network (BOTAN) [25]. Intriguingly, even with the same overall architecture, BOTAN finds a better prediction for the single particle dynamics, showing that the extra edge information improves the performance of the GNN. Pezzicoli *et al.* explored an alternative improvement by explicitly requiring the GNN to enforce the rotational symmetry, an idea sometimes referred to as geometric deep learning or rotation equivariant network (SE(3), [28]). This adaptation also improved on the original work of Ref. [21] over nearly all considered time scales.

In addition to the development of increasingly sophisticated ML methods to predict the dynamic propensity, significant efforts have been made to better capture important local features of the structural input. Shortly after the seminal work on GNNs [21], Boattini *et al.* incorporated the recursive updating properties of GNNs into a set of locally coarse-grained structural descriptors. They found that fitting just three generations of descriptors with a linear regression algorithm was sufficient to essentially reach the accuracy of the GNN — leading to a much simpler and more interpretable algorithm for fitting glassy dynamics. Interestingly, learning the dynamic propensity using these descriptors with non-linear models (MLP and GNN) did not improve the ability to predict the dynamics, as quantified by the Pearson correlation coefficient [24].

This approach has been further developed by including physics-inspired descriptors which have been identified as important structural proxies in the past thirty years of glass research (GlassMLP, [30]). These additional structural descriptors include potential energy and properties of the Voronoi cells, as well as the choice of describing the system in terms of its inherent state [30]. The inherent state corresponds to the energy minimum configuration that is closest to the actual input structure. These modifications improve the performance of the network. GlassMLP further uses MLP for supervised learning, which enables the precise representation of non-linear or non-Gaussian features such as probability distributions of propensities [30]. Using transferability in system size, the network has been applied to determine dynamic correlation lengths and the geometry of rearranging clusters over a wide range of temperatures. The GlassMLP model was recently enhanced to improve the transferability across time scales and temperatures and to explore physical regimes where direct training cannot be performed [89]. In a vein similar to GlassMLP, structural descriptors have been improved by going beyond inherent states and using cage states (CAGE, [26]). Cage states are extracted from restricted ensemble averages of the local structure using Monte Carlo simulations and thus better describe the local environment, which also helps improving the performance of the model.

By construction, all methods perform best at the temperature at which they are trained. However, it is pos-

sible to apply a trained network to other temperatures, and test how predictions correlate with true dynamics there [20, 21, 28, 83, 89]. Good performance in such transfer experiments indicates that the model captures relevant universal features in the structure-dynamics relationship of glass-forming liquids. Transferability has, for example, been used in Ref. [83] to investigate links between amorphous structure and fragility and in Ref. [89] to predict features of dynamic heterogeneity for temperatures comparable to the experimental glass transition temperature. In Sec. V, we show additional transferability experiments for several models.

In future work, it will be interesting to further exploit the transferability of supervised ML techniques to robustly extract information on structural relaxation at very low temperatures. One particular goal would concern the evolution of dynamic heterogeneity upon approaching the glass transition, and make the connection with experimental results [90]. One possible strategy to improve transferability of trained models to low-temperature regimes in which dynamics cannot be run for long enough (i.e., no or little labeled data is available) would be to use self-supervised learning [91]. More generally, self-supervised learning could also be used to enhance performance of the deep approaches (see also Sec. VI B).

To better capture the physical phenomena underpinning glassy dynamics, it seems necessary that the ML models not only faithfully predict the dynamics at the single particle level, but also correctly reproduce all statistical features of the propensity field, including spatial and temporal correlations, such as those measured by the four-point susceptibility $\chi_4(t)$ [92]. For GNNs, a known problem is over-smoothing, i.e., the predicted propensity field tends to be smoother than the ground truth. Jung *et al.* [30] introduced additional correlations-related terms in the loss function to avoid over-smoothing, and could show that the predictions indeed display more realistic correlation functions, even in a rather simple MLP architecture. An open direction for future works is the development of such improved loss functions, which could also be used in deep architectures.

A related idea is to learn not only the isoconfigurational average of the displacement, $\langle \Delta \mathbf{r}_i(t) \rangle_{\rm iso}$, but its full statistical distribution, $P_{\rm iso}(\Delta \mathbf{r}_i(t))$. This could be accomplished either by additionally fitting higher moments of the distribution, or by taking a generative model approach, in which one would attempt to generate realistic single-instance dynamical fields (not averaged in the isoconfigurational ensemble). This task could be achieved using a variational auto-encoder approach [93]. The spatial correlations of the isoconfigurational average being a priori different from the spatial correlations of single instances, the hope is that such generative models would reproduce these statistics more faithfully than conventional ones. To train such models, one should probably use single-instance configurations, which can also be used to regress the average. Finally, this approach might then be usable to propose new configurations on the structural relaxation time scale τ_α, with the long-term goal to be able to develop a Monte Carlo algorithm which completely avoids the critical slowing down of the dynamics on the approach to the glass transition. Similarly, the predictions could be used to create ultra-stable glasses by generating prototypical hard neighborhoods and remove structural "defects" [94]. In Sec. VI C we discuss in more detail generative models and their applications for sampling low-temperature glassy structures.

In physics, explaining complex behavior builds on the ability of theories and models to substantially compress the inherent information of a natural phenomenon [95]. From this standpoint, large machine learning models, involving several hundreds or thousands of directly fitted parameters, do not qualify as a physical theory in the traditional sense of the term. Moreover, due to their non-linearity, neural-networks models are still rather difficult to interpret, although some progress is being made [96, 97]. This does not mean, however, that large "black-box" ML models are useless in this context: their predictions can be instrumental as part of an eurisitic process, eventually leading to a simple solution to an outstanding problem [98]. This is nicely demonstrated by the results presented in this section: building on the insight of Ref. [21], Filion and coworkers achieved about the same prediction accuracy as graph neural networks using a much simpler and transparent linear regression method [24]. Linear models thus retain a strong appeal for fundamental research in glass physics, because of their simplicity and their direct mapping to the underlying structural descriptor. Interpreting the outcome of machine learning models also hinges on the ability of identifying the most relevant features – a process known as "feature selection". Analysis of the so-called "information imbalance" has emerged as a general and elegant approach to feature selection [99]. This method has been very recently applied to identify the most relevant structural features for glassy dynamics [100].

Finally, a natural aim for future investigations is to enlarge the ML studies described above to encompass diverse glass-forming materials with complex dynamics, including active glasses as model for biological tissues [101, 102]. Although it is generally accepted that equilibrium microscopic dynamics do not influence long-time structural relaxation it is unclear which signature activity plays on structural descriptors in active glasses [103]. Similarly, very little information exists on the dynamical properties of glasses during aging [104], and whether similarly strong structure-dynamics relationships can be found as for equilibrium relaxation.

IV. MACHINE LEARNING PHENOMENOLOGICAL GLASS MODELS

In this section, we demonstrate the potential of combining ML methods with physical insights to develop ef-